time. Road link availability is assumed to be both scenario-dependent and time-dependent. Rather than modeling each road link as either fully available or completely disrupted, this study represents the effective availability of each link by a continuous capacity proportion between 0 and 1. This formulation captures partial blockages, limited accessibility, and progressive road clearance operations after a disaster.

Assumption 3: Time- and scenario-dependent hospital receiving capacity: The receiving capacity of each hospital is assumed to vary across time periods and disaster scenarios. This reflects the operational uncertainty faced by hospitals after a disaster, including staff availability, medical resource depletion, facility damage, and gradual restoration of service capacity. Hospital receiving capacities are treated as exogenous uncertain parameters in the second-stage response problem.

Assumption 4: Within-period transportation approximation: Each discrete planning period is assumed to be sufficiently long relative to the travel time required on operational road links between disaster areas and CCPs, and between CCPs and hospitals. Therefore, casualty transportation and hospital transfer decisions are modeled as aggregate flows that can be completed within the same decision period. The effects of transportation difficulty are captured through road capacity availability and transportation cost or travel-time penalty parameters, without explicitly tracking en-route vehicles or inter-period travel states.

3.3 Sets, Parameters, and Decision Variables

Sets:

I	Set of disaster areas, indexed by i.
J	Set of candidate CCP locations, indexed by j.
H	Set of hospitals, indexed by h.
L	Set of casualty severity levels, indexed by l. Specifically, it includes minor, moderate, and severe levels.
L^{Amb}	Subset of casualty severity levels that require ambulance transportation and hospital transfer (i.e., moderate and severe casualties), $L^{Amb} \subseteq L$.
T	Set of discrete time periods, indexed by t.
S	Set of post-disaster scenarios, indexed by s.

First-Stage Deterministic Parameters:

f_j	Fixed cost of opening CCP j, $\forall j \in J$.
cv	Unit cost of assigning one medical staff member to a CCP.
ca	Unit cost of assigning one CCP ambulance to a CCP.

cy_{hj}	Unit cost of allocating one unit of medical supplies from hospital h to CCP j, $\forall j \in J$, $\forall h \in H$.
nv	Total number of available medical staff members.
na	Total number of available CCP ambulances.
$\bar{s}_h$	Maximum amount of medical supplies that hospital h can provide, $\forall h \in H$.
$\bar{v}_j$	Maximum number of medical staff members that can be assigned to CCP j, $\forall j \in J$.
$\bar{u}_j$	Maximum number of CCP ambulances that can be assigned to CCP j, $\forall j \in J$.
$\bar{y}_j$	Maximum amount of medical supplies that can be assigned to CCP j , $\forall j \in J$.

Second-Stage Deterministic Parameters:

c_{ij}	Normal maximum transportation capacity (casualties/period) on the road link from disaster area i to CCP j, $\forall i \in I, j \in J$.
c_{jh}	Normal maximum transportation capacity (casualties/period) on the road link from CCP j to hospital h, $\forall j \in J, h \in H$.
κ	Number of ambulance-requiring casualties that one CCP ambulance can transport in one period.
η	Number of casualties that one hospital ambulance can transfer in one period.
b_h	Number of ambulances owned by hospital h, $\forall h \in H$.
k_{jl}	Physical capacity of CCP j for casualties of severity level l, $\forall j \in J, l \in L$.
τ_l	Number of periods required to treat one casualty of severity level l at a CCP, $\forall l \in L$.
α_l	Number of casualties of severity level \$l\$ that one medical staff member can treat in one period, $\forall l \in L$.
β_l	Amount of medical supplies consumed by one casualty of severity level l, $\forall l \in L$.
ρ_l	Unit penalty cost for one casualty of severity level l remaining in a disaster area for one period, $\forall l \in L$.
δ_l	Unit penalty cost for one treated casualty of severity level $\forall l \in L^{Amb}$ waiting at a CCP for hospital transfer for one period, $\forall l \in L^{Amb}$.
t_{ij}	Unit transportation cost, or cost-equivalent travel time penalty, for transporting one casualty from disaster area i to CCP j, $\forall i \in I, j \in J$.
t_{jh}	Unit transportation cost, or cost-equivalent travel time penalty, for transferring one casualty

from CCP j to hospital h, $\forall j \in J, h \in H$.

Random Variables:

$\tilde{\xi}_{ilt}$ Random variable representing the number of newly generated casualties of severity level l in disaster area i during period t. Its realization in scenario s is ξ_{ilts}.

$\tilde{u}_{ijt}$ Random variable representing the available capacity proportion ([0,1]) of the road link from disaster area i to CCP j during period t. Its realization in scenario s is u_{ijts}.

$\tilde{w}_{jht}$ Random variable representing the available capacity proportion ([0,1]) of the road link from CCP j to hospital h during period t. Its realization in scenario s is w_{jhts}.

$\tilde{h}_{ht}$ Random variable representing the maximum number of casualties that hospital h can receive during period t. Its realization in scenario s is h_{hts}.

First-Stage Decision Variables:

X_j $X_j \in \{0,1\}$,1 if CCP j is opened, and 0 otherwise, $\forall j \in J$.

V_j Number of medical staff members assigned to CCP j, $\forall j \in J$.

U_j Number of CCP ambulances assigned to CCP j, $\forall j \in J$.

Y_{hj} Amount of medical supplies allocated from hospital h to CCP j, $\forall j \in J, h \in H$.

Second-Stage Decision Variables:

$FI_{ijlt}(s)$ Number of casualties of severity level l transported from disaster area i to CCP j during period t under scenario s, $\forall i \in I, j \in J, l \in L, t \in T$.

$FO_{jhlt}(s)$ Number of casualties of severity level $l \in L^{Amb}$ transferred from CCP j to hospital h during period t under scenario s, $\forall j \in J, h \in H, l \in L^{Amb}, t \in T$.

$RM_{ilt}(s)$ Number of casualties of severity level l remaining in disaster area i at the end of period t under scenario s, $\forall i \in I, l \in L, t \in T$.

$REG_{jlt}(s)$ Number of casualties of severity level l registered at CCP j during period t under scenario s, $\forall j \in J, l \in L, t \in T$.

$TRT_{jlt}(s)$ Number of casualties of severity level l under treatment at CCP j during period t under scenario s, $\forall j \in J, l \in L, t \in T$.

$WAT_{jlt}(s)$ Number of treated casualties of severity level $l \in L^{Amb}$ waiting at CCP j for hospital transfer at the end of period t under scenario s, $\forall j \in J, l \in L^{Amb}, t \in T$.

3.4 Risk-averse two-stages stochastic programming

The first-stage objective is to minimize the total pre-disaster preparation cost and the expected post-disaster response cost:

$$\min \quad \sum_{j\in J} f_j X_j + cv\sum_{j\in J} V_j + ca\sum_{j\in J} U_j + \sum_{h\in H}\sum_{j\in J} c\,y_{hj}Y_{hj} + \mathbb{E}_{\tilde{\omega}}[Q(X,V,U,Y;\tilde{\omega})] \quad (1)$$

The first-stage decisions are subject to the following constraints:

$$\sum_{j\in J} V_j \le nv \quad (2)$$

$$\sum_{j\in J} U_j \le na \quad (3)$$

$$\sum_{j\in J} Y_{hj} \le \bar{s}_h, \qquad \forall h \in H \quad (4)$$

$$V_j \le \bar{v}_j X_j, \qquad \forall j \in J \quad (5)$$

$$U_j \le \bar{u}_j X_j, \qquad \forall j \in J \quad (6)$$

$$\sum_{h\in H} Y_{hj} \le \bar{y}_j X_j, \qquad \forall j \in J \quad (7)$$

$$X_j \in \{0,1\}, \qquad \forall j \in J \quad (8)$$

$$V_j, U_j, Y_{hj} \in \mathbb{Z}_+, \qquad \forall j \in J, h \in H \quad (9)$$

Constraint (2) ensures that the total number of medical staff members assigned to all opened CCPs does not exceed the available number of medical staff. Constraint (3) limits the total number of CCP ambulances assigned to all CCPs by the available CCP ambulance fleet. Constraint (4) ensures that the amount of medical supplies allocated from each hospital does not exceed its available supply. Constraint (5) allows medical staff to be assigned to CCP j only if CCP j is opened. Constraint (6) allows CCP ambulances to be assigned to CCP j only if CCP j is opened. Constraint (7) ensures that medical supplies can be allocated to CCP j only when CCP j is opened. Constraints (8) and (9) define the domains of the first-stage decision variables.

Given the first-stage decisions (X, v, θ, y) and the realization of scenario $\boldsymbol{\omega}^s$, the second-stage recourse function is defined as follows:

$$Q(X,V,U,Y;\omega_s) = \min \quad \sum_{t\in T}\sum_{i\in I}\sum_{l\in L}\rho_l\, RM_{ilt}(s) + \sum_{t\in T}\sum_{j\in J}\sum_{l\in L^{Amb}}\delta_l\, WAT_{jlt}(s) + \sum_{t\in T}\sum_{i\in I}\sum_{j\in J}\sum_{l\in L} t_{ij}\, FI_{ijlt}(s) + \sum_{t\in T}\sum_{j\in J}\sum_{h\in H}\sum_{l\in L^{Amb}} t_{jh}\, FO_{jhlt}(s) \tag{10}$$

s.t.

$$\sum_{l\in L} FI_{ijlt}(s) \le c_{ij}u_{ijts}X_j, \qquad \forall i\in I, j\in J, t\in T, s\in S \tag{11}$$

$$\sum_{l\in L^{Amb}} FO_{jhlt}(s) \le c_{jh}w_{jhts}X_j, \qquad \forall j\in J, h\in H, t\in T, s\in S \tag{12}$$

$$\sum_{i\in I}\sum_{l\in L^{Amb}} FI_{ijlt}(s) \le \kappa U_j, \qquad \forall j\in J, t\in T, s\in S \tag{13}$$

$$\sum_{j\in J}\sum_{l\in L^{Amb}} FO_{jhlt}(s) \le \eta b_h, \qquad \forall h\in H, t\in T, s\in S \tag{14}$$

$$RM_{ilt}(s) = RM_{il,t-1}(s) + \xi_{ilts} - \sum_{j\in J} FI_{ijlt}(s), \forall i\in I, l\in L, t\in T, s\in S \tag{15}$$

$$RM_{il0}(s) = 0, \qquad \forall i\in I, l\in L, s\in S \tag{16}$$

$$REG_{jlt}(s) = \sum_{i\in I} FI_{ijlt}(s), \qquad \forall j\in J, l\in L, t\in T, s\in S \tag{17}$$

$$TRT_{jlt}(s) = \sum_{r=\max(1,t-\tau_l+1)}^{t} REG_{jlr}(s), \qquad \forall j\in J, l\in L, t\in T, s\in S \tag{18}$$

$$WAT_{jlt}(s) = WAT_{jl,t-1}(s) + REG_{jl,t-\tau_l}(s) - \sum_{h\in H} FO_{jhlt}(s) \quad , j\in J, l\in L^{Amb}, t\in T, s\in S \tag{19}$$

$$WAT_{jl0}(s) = 0, \qquad \forall j\in J, l\in L^{Amb}, s\in S \tag{20}$$

$$REG_{jl,t-\tau_l}(s) = 0 \quad \text{if } t-\tau_l < 1 \tag{21}$$

$$TRT_{jlt}(s) + WAT_{jlt}(s) \le k_{jl}X_j, \qquad \forall j\in J, l\in L^{Amb}, t\in T, s\in S \tag{22}$$

$$TRT_{jlt}(s) \le k_{jl}X_j, \qquad \forall j\in J, l\in L\setminus L^{Amb}, t\in T, s\in S \tag{23}$$

$$\sum_{l\in L}\frac{TRT_{jlt}(s)}{\alpha_l} \le V_j, \qquad \forall j\in J, t\in T, s\in S \tag{24}$$

$$\sum_{t\in T}\sum_{l\in L}\beta_l\, REG_{jlt}(s) \le \sum_{h\in H} Y_{hj}, \qquad \forall j\in J, s\in S \tag{25}$$

$$\sum_{j\in J}\sum_{l\in L^{Amb}} FO_{jhlt}(s) \le h_{hts}, \qquad \forall h\in H, t\in T, s\in S \tag{26}$$

$$FI_{ijlt}(s), FO_{jhlt}(s), RM_{ilt}(s), REG_{jlt}(s), TRT_{jlt}(s), WAT_{jlt}(s) \ge 0, \tag{27}$$

$$\forall i \in I,\ j \in J,\ h \in H,\ l \in L,\ t \in T,\ s \in S$$

The first term in (10) penalizes casualties remaining in disaster areas. The second term penalizes treated ambulance-requiring casualties waiting at CCPs for hospital transfer. The third term represents the transportation cost, or travel-time penalty, for moving casualties from disaster areas to CCPs. The fourth term represents the transportation cost, or travel-time penalty, for transferring ambulance-requiring casualties from CCPs to hospitals.

Constraints (11) and (12) capture the dynamic and partial availability of transportation links after a disaster. Constraint (13) limits the number of ambulance-requiring casualties transported from disaster areas to CCP j by the number of CCP ambulances assigned to CCP j. Constraint (14) limits the number of ambulance-requiring casualties transferred from CCPs to hospital h by the hospital ambulance fleet. Constraint (15) tracks the number of casualties remaining in each disaster area at the end of each period. Constraint (16) specifies the initial condition. Constraint (17) states that the number of casualties registered at CCP j equals the total number of casualties transported to CCP j from all disaster areas during the same period. Constraint (18) represents the number of casualties of severity level l that are under treatment at CCP j during period t. A casualty remains under treatment for τ_l periods after registration. Constraint (19) tracks the number of treated ambulance-requiring casualties waiting at CCP j for hospital transfer. Constraint (20) defines the initial waiting condition, and constraint (21) specifies the boundary condition for treatment completion. Constraints (22)–(23) ensure that CCP capacity is respected at both the severity-specific level and the total facility level. Constraint (24) ensures that the treatment workload at each CCP does not exceed the treatment capacity provided by the assigned medical staff. Constraint (25) ensures that the total amount of medical supplies consumed by casualties registered at CCP j does not exceed the amount of supplies allocated to that CCP. Constraint (26) limits the number of casualties transferred to each hospital by the scenario-specific hospital receiving capacity in each period. Constraint (27) defines the non-negativity domains of the second-stage decision variables. For variables that are only defined for $l \in L^{Amb}$, such as $F^{OUT}_{jhlt}(s)$ and $Wait_{jlt}(s)$, the corresponding severity index is restricted to L^{Amb}.

3.5 Distributionally robust optimization model

3.5.1 Ambiguity Sets

Definition 1: box set